

2.3 ChIP-seq

徐英泽 2024011267

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1. 请解释在ChIP-seq实验中为什么一般都要平行做一个 control（通常叫 input）的实验。

- Input 是将染色质超声打断后，不经过任何抗体免疫沉淀，直接纯化、建库并测序的样品。它代表了样本在抗体富集步骤前的总 DNA 组成。
- 由于基因组并非完全均一，某些区域天然更容易被物理打断、不容易扩增或在测序中产生偏向性，这些都会造成某些区域背景信号异常升高。如果不排除这些区域，软件可能将这种实验噪音误判为转录因子结合或组蛋白修饰的真实信号。

2. 请解释 findPeaks 和 findMotifsGenome.pl 主要参数的含义。

- **findPeaks:**
 - `-style`: 分析模式，包括 `factor` / `histone` 模式。
 - `-o`: 输出文件路径。
 - `-i`: 存储输入样本中间文件的“tag directory”。
 - `-size`: Peak 大小。在此窗口内计算 tag 密度来寻找局部最大值。
 - `-minDist`: 峰之间的最小距离，防止在同一个结合位点附近重复找峰。
 - `-fdr`: 错误发现率阈值。
 - `gsize`: 有效基因组大小。
 - `center`: 中心化，分析结束后将峰的位置移动到 tag 堆积密度的最高点。
- **findMotifsGenome.pl:**
 - `-size`: 用于分析的区域大小，决定从 Peak 的中心向两边各取多少 bp 的序列进行分析。
 - `len`: Motif 长度，指定需要寻找的基序长度，允许同时输入多个长度。
 - `genome`: 参考基因组版本，指定分析所用的基因组。

3. 我们在容器的/home/test/chip-seq/homework目录中提供了酵母Snf1蛋白CHIP-seq的bam文件，ip.chrom_part.bam为IP实验数据，input.chrom_part.bam为背景数据。请大家从这两个文件出发，用homer重复本章中介绍的peak calling和motif finding分析。请大家提交找到的motif的截图，以及Fold Change (vs Control) ≥ 8 且p-value (vs Control) $< 10^{-8}$ 的peaks。

```
PS C:\Users\54727> docker exec -it -u root XuYingze_Linux bash
root@bioinfo_docker:/home/test/chip-seq/homework# mkdir output

# 使用makeTagDirectory命令产生peak calling所需的中间文件。
root@bioinfo_docker:/home/test/chip-seq# makeTagDirectory homework/ip
homework/ip.chrom_part.bam
root@bioinfo_docker:/home/test/chip-seq# makeTagDirectory homework/input
homework/input.chrom_part.bam

# 从这两组中间文件出发进行peak calling。
root@bioinfo_docker:/home/test/chip-seq# findPeaks homework/ip/ -style factor -o
homework/output/part.peak -i homework/input/

# 简单查看part.peak文件。
root@bioinfo_docker:/home/test/chip-seq/homework/output# head -50 part.peak
# HOMER Peaks
# Peak finding parameters:
# tag directory = homework/ip/
#
# total peaks = 69
# peak size = 248
# peaks found using tags on both strands
# minimum distance between peaks = 496
# fragment length = 232
# genome size = 2097191
# Total tags = 1402413.0
# Total tags in peaks = 135829.0
# Approximate IP efficiency = 9.69%
# tags per bp = 0.668669
# expected tags per peak = 165.830
# maximum tags considered per bp = 66.0
# effective number of tags used for normalization = 10000000.0
# Peaks have been centered at maximum tag pile-up
# FDR rate threshold = 0.001000000
# FDR effective poisson threshold = 1.876888e-05
# FDR tag threshold = 222.0
# number of putative peaks = 402
#
# input tag directory = homework/input/
# Fold over input required = 4.00
# Poisson p-value over input required = 1.00e-04
# Putative peaks filtered by input = 323
#
```

```
# size of region used for local filtering = 10000
# Fold over local region required = 4.00
# Poisson p-value over local region required = 1.00e-04
# Putative peaks filtered by local signal = 10
#
# Maximum fold under expected unique positions for tags = 2.00
# Putative peaks filtered for being too clonal = 0
#
# cmd = findPeaks homework/ip/ -style factor -o homework/output/part.peak -i homework/input/
#
# Column Headers:
#PeakID chr      start  end      strand Normalized Tag Count      focus ratio      findPeaks
Score      Total Tags      Control Tags (normalized to IP Experiment)      Fold Change vs
Control p-value vs Control      Fold Change vs Local      p-value vs Local      Clonal Fold
Change
chrIV-1 chrIV    465220 465468 +        111129.9      0.920 15510.000000 15585.0234.1
66.57 0.00e+00      55.11 0.00e+00      0.50
chrIV-2 chrIV    1490100 1490348 +        81687.8 0.857 11468.000000 11456.0 195.1 58.72
0.00e+00 35.06 0.00e+00      0.50
chrV-1 chrV      141138 141386 +        54449.0 0.855 7647.000000 7636.0 182.3 41.88
0.00e+00 21.55 0.00e+00      0.52
chrV-2 chrV      69078 69326 +        48659.0 0.823 6837.000000 6824.0 206.5 33.05
0.00e+00 20.52 0.00e+00      0.50
chrV-3 chrV      85195 85443 +        46277.4 0.861 6493.000000 6490.0 225.6 28.77
0.00e+00 21.56 0.00e+00      0.50
chrIV-3 chrIV    1080509 1080757 +        34405.0 0.832 4830.000000 4825.0 234.1 20.61
0.00e+00 23.11 0.00e+00      0.50
chrIV-4 chrIV    599953 600201 +        26597.0 0.755 3733.000000 3730.0 190.1 19.62
0.00e+00 15.58 0.00e+00      0.50
chrV-4 chrV      321939 322187 +        24821.5 0.754 3484.000000 3481.0 177.4 19.63
0.00e+00 13.66 0.00e+00      0.50
chrIV-5 chrIV    1468786 1469034 +        23595.0 0.794 3317.000000 3309.0 193.7 17.08
0.00e+00 14.36 0.00e+00      0.51
chrIV-6 chrIV    132817 133065 +        19402.3 0.782 2723.000000 2721.0 209.3 13.00
0.00e+00 11.95 0.00e+00      0.52

# 使用findMotifsGenome.pl进行motif finding。
root@bioinfo_docker:/home/test/chip-seq# findMotifsGenome.pl homework/output/part.peak
sacCer2 homework/output/part.motif.output -len 8

# 将output复制到宿主机，并查看html文件
PS C:\Users\54727> docker cp XuYingze_Linux:/home/test/chip-
seq/homework/output/part.motif.output/ "C:\Users\54727\Downloads"
Successfully copied 988kB to C:\Users\54727\Downloads
```

motif的截图如下:

Homer *de novo* Motif Results (homework/output/part.motif.output/)

[Known Motif Enrichment Results](#)

[Gene Ontology Enrichment Results](#)

If Homer is having trouble matching a motif to a known motif, try copy/pasting the matrix file into [STAMP](#)

More information on motif finding results: [HOMER](#) | [Description of Results](#) | [Tips](#)

Total target sequences = 69

Total background sequences = 34749

* - possible false positive

Rank	Motif	P-value	log P-value	% of Targets	% of Background	STD(Bg STD)	Best Match/Details	Motif File
1		1e-18	-4.267e+01	36.23%	3.47%	35.9bp (103.2bp)	OPI1/Literature(Harbison)/Yeast(0.938) More Information Similar Motifs Found	motif file (matrix)
2 *		1e-11	-2.623e+01	65.22%	25.40%	61.0bp (94.7bp)	AZF1/MA0277.1/Jaspar(0.831) More Information Similar Motifs Found	motif file (matrix)
3 *		1e-10	-2.406e+01	42.03%	10.92%	48.9bp (120.9bp)	PDR3/MA0353.1/Jaspar(0.799) More Information Similar Motifs Found	motif file (matrix)
4 *		1e-5	-1.270e+01	11.59%	1.28%	50.6bp (87.2bp)	UME6/UME6_YPD/51-UME6(Harbison)/Yeast(0.697) More Information Similar Motifs Found	motif file (matrix)
5 *		1e-5	-1.240e+01	14.49%	2.30%	44.9bp (82.1bp)	POL008.1_DCE_S_1/Jaspar(0.685) More Information Similar Motifs Found	motif file (matrix)
6 *		1e-4	-1.104e+01	15.94%	3.31%	58.3bp (91.0bp)	RPH1/MA0372.1/Jaspar(0.712) More Information Similar Motifs Found	motif file (matrix)
7 *		1e-4	-1.076e+01	71.01%	45.90%	49.6bp (148.1bp)	RSC30/MA0375.1/Jaspar(0.834) More Information Similar Motifs Found	motif file (matrix)
8 *		1e-3	-8.015e+00	8.70%	1.34%	39.3bp (118.0bp)	RTG3(MacIsaac)/Yeast(0.656) More Information Similar Motifs Found	motif file (matrix)
9 *		1e-1	-4.115e+00	8.70%	2.96%	40.2bp (135.8bp)	AFT2/MA0270.1/Jaspar(0.811) More Information Similar Motifs Found	motif file (matrix)

寻找并输出Fold Change (vs Control) >=8且p-value (vs Control) < 10⁻⁸的peaks。

```
root@bioinfo_docker:/home/test/chip-seq/homework/output# grep -v '#' part.peak | awk '$11>=8
&& $12<1/10^8{print $0}'
```

```
chrIV-1 chrIV 465220 465468 + 111129.9 0.920 15510.000000 15585.0234.1
66.57 0.00e+00 55.11 0.00e+00 0.50
chrIV-2 chrIV 1490100 1490348 + 81687.8 0.857 11468.000000 11456.0 195.1 58.72
0.00e+00 35.06 0.00e+00 0.50
chrV-1 chrV 141138 141386 + 54449.0 0.855 7647.000000 7636.0 182.3 41.88
0.00e+00 21.55 0.00e+00 0.52
chrV-2 chrV 69078 69326 + 48659.0 0.823 6837.000000 6824.0 206.5 33.05
0.00e+00 20.52 0.00e+00 0.50
chrV-3 chrV 85195 85443 + 46277.4 0.861 6493.000000 6490.0 225.6 28.77
0.00e+00 21.56 0.00e+00 0.50
chrIV-3 chrIV 1080509 1080757 + 34405.0 0.832 4830.000000 4825.0 234.1 20.61
0.00e+00 23.11 0.00e+00 0.50
chrIV-4 chrIV 599953 600201 + 26597.0 0.755 3733.000000 3730.0 190.1 19.62
0.00e+00 15.58 0.00e+00 0.50
chrV-4 chrV 321939 322187 + 24821.5 0.754 3484.000000 3481.0 177.4 19.63
0.00e+00 13.66 0.00e+00 0.50
chrIV-5 chrIV 1468786 1469034 + 23595.0 0.794 3317.000000 3309.0 193.7 17.08
0.00e+00 14.36 0.00e+00 0.51
chrIV-6 chrIV 132817 133065 + 19402.3 0.782 2723.000000 2721.0 209.3 13.00
0.00e+00 11.95 0.00e+00 0.52
chrIV-7 chrIV 591669 591917 + 18304.2 0.792 2568.000000 2567.0 200.1 12.83
0.00e+00 12.88 0.00e+00 0.50
chrIV-8 chrIV 721812 722060 + 17840.7 0.739 2514.000000 2502.0 192.3 13.01
0.00e+00 9.61 0.00e+00 0.51
chrIV-9 chrIV 1 230 + 17206.1 0.884 2444.000000 2430.0 60.3 40.30
0.00e+00 632.81 0.00e+00 0.81
```

chrIV-10	chrIV	1233763	1234011	+	16250.6	0.888	2303.000000	2279.0156.1
14.60	0.00e+00	10.68	0.00e+00		0.52			
chrIV-11	chrIV	234340	234588	+	16179.3	0.790	2275.000000	2269.0199.4
11.38	0.00e+00	9.37	0.00e+00		0.51			
chrV-5	chrV	225453	225701	+	15066.9	0.901	2123.000000	2113.0 157.5 13.42
0.00e+00	11.73	0.00e+00		0.53				
chrIV-12	chrIV	357166	357414	+	15052.6	0.705	2113.000000	2111.0205.7
10.26	0.00e+00	8.19	0.00e+00		0.51			
chrIV-13	chrIV	416932	417180	+	13954.5	0.838	1973.000000	1957.0188.0
10.41	0.00e+00	10.83	0.00e+00		0.51			
chrIV-15	chrIV	1278678	1278926	+	13697.8	0.852	1925.000000	1921.0225.6
8.51	0.00e+00	12.86	0.00e+00		0.52			
chrIV-16	chrIV	1164971	1165219	+	13419.7	0.746	1887.000000	1882.0215.7
8.73	0.00e+00	9.29	0.00e+00		0.52			
chrV-6	chrV	491091	491339	+	12457.1	0.773	1749.000000	1747.0 180.9 9.66
0.00e+00	13.69	0.00e+00		0.52				
chrIV-14	chrIV	1525285	1525496	+	11779.7	0.923	1953.000000	1652.058.2
28.40	0.00e+00	32.51	0.00e+00		1.27			
chrIV-17	chrIV	722439	722687	+	10774.3	0.676	1512.000000	1511.0172.4
8.76	0.00e+00	5.26	0.00e+00		0.53			
chrIV-46	chrIV	568825	569073	+	5005.7	0.820	705.000000	702.0 85.8
8.18	0.00e+00	4.54	3.86e-226		0.84			